*Davriy jadvaldan foydalanish mumkin.*

**1.** $2SO_2 + O_2 \rightleftarrows 2SO_3$ reaksiyasi 4 l hajmli idishda o‘tkazildi. Ma’lumotlar jadvalda berilgan:

| Modda | $SO_2$ | $O_2$ | $SO_3$ |
|---|---|---|---|
| Boshlang‘ich, mol | 6 | 4 | 0 |
| Muvozanatda, mol | — | — | 4 |

$SO_2$ ning necha foizi reaksiyaga **kirishganini aniqlang.**
A) *33,3* B) *66,7* C) *100* D) *133,3*

**2.** 16,4 l $CO_2$ -23 °C va 0,5 atm bosimda necha gramm keladi? ($R = 0{,}082$ l·atm/(mol·K))
A) *35,2* B) *8,8* C) *17,6* D) *32,21*

**3.** Quyidagi moddalarning qaysilarida donor–akseptor bog‘ mavjud? 1) $CO_2$; 2) $KNO_3$; 3) NaOH; 4) $(NH_4)_2SO_4$
A) *3, 4* B) *2, 4* C) *2, 3* D) *4*

**4.** $N_2O_5$ moddaning parchalanish reaksiyasida konsentratsiyaning vaqt bo‘yicha o‘zgarishi jadvalda berilgan:

| t, min | c, mol/l |
|---|---|
| 0 | 2 |
| 2 | 1,7 |
| 4 | 1,4 |
| 6 | 1,1 |

0-minutdan 2-minutgacha o‘rtacha tezlikni mol/(l·min) **hisoblang.**
A) *0,3* B) *0,075* C) *1* D) *0,15*

**5.** Faqat qaytaruvchi bo‘la oladigan **ionlarni aniqlang**: 1) $Cl^-$; 2) $Fe^{3+}$; 3) $Fe^{2+}$; 4) $K^+$
A) *2, 3* B) *3, 4* C) *1, 4* D) *1*

**6.** 0,5 mol $CO_2$ tarkibidagi atomlar **sonini toping** ($N_A = 6{,}02\cdot10^{23}$). Javobni $10^{23}$ birligida yozing.
A) *1,806* B) *0,451* C) *0,903* D) *0,301*

**7.** Hajmi 2 l bo‘lgan idishda A modda miqdori 30 s davomida 6 mol ga kamaydi. Reaksiyaning o‘rtacha tezligini (mol/(l·s)) **toping.**
A) *0,1* B) *0,2* C) *3* D) *0,4*

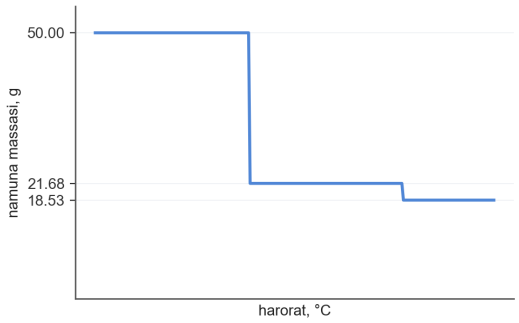
**8.** 0,1 M $H_3PO_4$ eritmasining normal **konsentratsiyasini (n) aniqlang.**
A) *0,1* B) *0,6* C) *0,3* D) *0,033*

**9.** $Zn + HNO_3 \rightarrow Zn(NO_3)_2 + NO + H_2O$ reaksiyasida 6 mol Zn erisa, $HNO_3$ ning necha moli oksidlovchi vazifasini bajaradi?
A) *16* B) *12* C) *8* D) *4*

**10.** $2SO_2 + O_2 \rightleftarrows 2SO_3$ reaksiyada muvozanat holatida $[SO_2] = 0{,}5$, $[O_2] = 0{,}5$, $[SO_3] = 4$ mol/l. Muvozanat **konstantasini (K) hisoblang.**
A) *256* B) *128* C) *64* D) *16*

**11.** Quyidagi tuzlarning qaysilari suvli eritmasi ishqoriy muhitga ega? 1) $NaNO_2$; 2) $CuSO_4$; 3) $NH_4Cl$; 4) NaCl
A) *1* B) *1, 2* C) *3* D) *4*

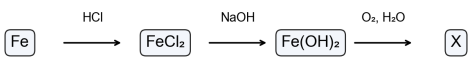
**12.** $Na_2CO_3\cdot10H_2O$ kristallogidrat namunasi (50 g) bosqichma-bosqich qizdirilganda uning massasi o‘zgarishi grafikda keltirilgan.




To‘liq suvsizlangandan keyin qolgan $Na_2CO_3$ ning **massasini (g) toping.**
A) *31,47* B) *21,68* C) *18,53* D) *37,06*

**13.** NaCl ning suvli eritmasi inert elektrodlar bilan elektroliz qilinganda qaysi moddalar ajraladi?
A) *Zn, $H_2$ va $O_2$* B) *Ag va $O_2$*
C) *Cu va $O_2$* D) *$H_2$, $Cl_2$ va NaOH*

**14.** Rasmdagi o‘zgarishlar zanjirida *X* moddaning **formulasini aniqlang.**




A) *$Fe(OH)_3$* B) *$FeCl_2$* C) *$SO_3$* D) *$Fe(OH)_2$*

**15.** Cu → ? → ? → ? o‘zgarishlar zanjirida (reagentlar: $O_2$, $H_2SO_4$, NaOH) 1-bosqichda hosil bo‘ladigan modda **formulasini toping.**
A) *$Na_2SO_3$* B) *$Ca(OH)_2$* C) *CuO* D) *NaCl*

**16.** $NH_3 + H_3PO_4 \rightarrow NH_4H_2PO_4$ jarayoni uchun 4,48 l (n.sh.) $NH_3$ olingan (unum 90%). $NH_4H_2PO_4$ **miqdorini (mol) aniqlang.**
A) *0,2* B) *0,36* C) *0,18* D) *0,09*

**17.** 14,8 g $Mg(NO_3)_2$ to‘liq parchalanguncha qizdirildi. Qattiq qoldiq **massasini (g) aniqlang.**
A) *4* B) *14,8* C) *8* D) *2*

**18.** $S + O_2 \rightarrow SO_2$ jarayoni uchun 8,96 l (n.sh.) $O_2$ olingan. $SO_2$ **hajmini (l, n.sh.) aniqlang.**
A) *17,92* B) *4,48* C) *8,96* D) *13,44*

**19.** $2NO + O_2 \rightarrow 2NO_2$ jarayoni uchun 8,96 l (n.sh.) $O_2$ olingan. NO **hajmini (l, n.sh.) aniqlang.**
A) *4,48* B) *17,92* C) *35,84* D) *8,96*

**20.** $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ jarayoni uchun 8,96 l (n.sh.) $C_3H_8$ olingan. $CO_2$ **hajmini (l, n.sh.) aniqlang.**
A) *2,99* B) *53,76* C) *26,88* D) *13,44*

**21.** 46 g $C_2H_5OH$ dan quyidagi ikki bosqichda $C_2H_4Cl_2$ olinadi: 1) $C_2H_5OH \rightarrow C_2H_4 + H_2O$ (unum 80%); 2) $C_2H_4 + Cl_2 \rightarrow C_2H_4Cl_2$ (unum 75%). Oxirgi mahsulot $C_2H_4Cl_2$ ning **massasini (g) toping.**
A) *59,4* B) *74,25* C) *79,2* D) *99*

**22.** Arenlar, neft va tabiiy gaz haqida qaysi fikrlar to‘g‘ri? 1) Neft fraksiyalaridan gazolin 150 °C gacha haydaladi; 2) Benzolda 6 ta $sp^2$ gibridlangan uglerod bor; 3) Benzoldan nitrobenzol nitrolash bilan olinadi; 4) Benzol uchun birikish reaksiyalari mutlaqo xos emas
A) *1, 2, 3* B) *1, 4* C) *2, 3* D) *2, 3, 4*

**23.** $C_2H_4 + H_2O$ ($H^+$) reaksiyasi natijasida qaysi organik modda hosil bo‘ladi?
A) *benzol* B) *asetilen* C) *etilen* D) *etanol*

**24.** Aldegid va ketonlar haqida qaysi fikrlar to‘g‘ri? 1) Ketonlar kuchsiz oksidlovchilar bilan oksidlanmaydi; 2) Sirka aldegid Kucherov reaksiyasi bilan olinadi; 3) Aldegidlar "kumush ko‘zgu" reaksiyasini beradi; 4) Formaldegid gaz holatda, suvda yaxshi eriydi
A) *1, 2, 4* B) *2, 3, 4*
C) *1, 3, 4* D) *1, 2, 3, 4*

**25.** $C_4H_8O_2$ karbon kislotalarning izomerlari soni nechta?
A) *3* B) *1* C) *4* D) *2*

**26.** 22,2 g propion kislota va 9,2 g etanol dan 80% unum bilan murakkab efir olindi. Efir **massasini (g) toping.**
A) *20,4* B) *16,32* C) *24,48* D) *32,64*

**27.** $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ reaksiyasida 44 g $CO_2$ ishtirok etdi. $O_2$ **hajmini (l, n.sh.) toping.**
A) *44,8* B) *22,4* C) *11,2* D) *33,6*

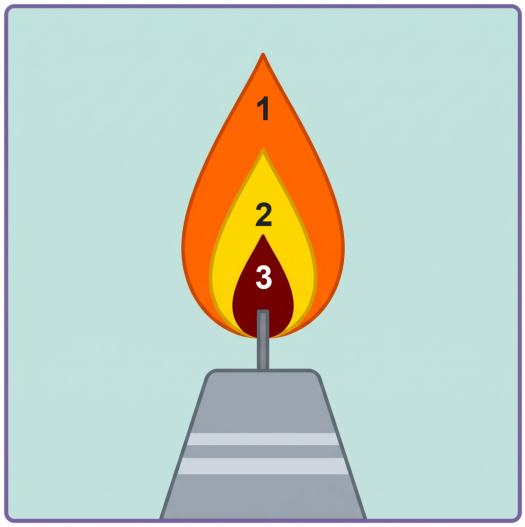
**28.** Aminlar, aminokislotalar va oqsillar haqida qaysi fikrlar to‘g‘ri? 1) Nitrobenzoldan anilin Zinin reaksiyasi bilan olinadi; 2) Aminlar kuchsiz asoslar xossasini namoyon qiladi; 3) Aminokislotalar faqat kislotali xossaga ega; 4) Dipeptidda 2 ta peptid bog‘ bor
A) *4* B) *1, 2* C) *1* D) *3*

**29.** Yuqori molekulyar birikmalar haqida qaysi fikrlar to‘g‘ri? 1) Lavsan poliamid tola; 2) Kapron poliamid tola; 3) Tabiiy kauchuk – sis-1,4-poliizopren; 4) Teflon — polivinilxlorid
A) *2, 3* B) *3, 4* C) *1, 2* D) *3*

**30.** 100 g 30% li va 100 g 40% li NaOH eritmalari aralashtirildi. Yangi eritmaning massa **ulushini (%) toping.**
A) *70* B) *35* C) *31,5* D) *38,5*

**31.** Kimyo laboratoriyasida ishlash qoidalari haqida qaysi fikrlar to‘g‘ri? 1) Ishqor teriga tushsa, qo‘l suv bilan yuviladi; 2) Probirka qizdirilganda og‘zi odamga qaratilmaydi; 3) Zaharli gaz ajraladigan tajriba mo‘rili shkafda o‘tkaziladi; 4) Konsentrlangan kislotani suvga ohista, aralashtirib quyiladi
A) *1, 2, 3, 4* B) *2, 3, 4*
C) *1, 3, 4* D) *1, 2, 4*

**32.** Rasmda spirt lampasi alangasi tasvirlangan. Alanganing qaysi qismi eng kam issiq va pilikka yaqin?




A) *hamma qismi bir xil* B) *3 – pastki qism*
C) *2 – o‘rta qism* D) *1 – yuqori qism* —

| | |
|---|---|
| **Topshiriqlar (33–35) va javob variant (A–F)larini o‘zaro to‘g‘ri moslashtiring.** Metan ($CH_4$) va propen ($C_3H_6$) dan iborat 7,84 l (n.sh.) aralashma bromli suvdan o‘tkazilganda 8 g brom yutildi. | A *88*<br>B *11,43*<br>C *21,43*<br>D *14,29*<br>E *1,12*<br>F *176* |
| **33. Aralashmadagi propen ($C_3H_6$) hajmini (l, n.sh.) aniqlang.** | |
| **34. Aralashmadagi propen ($C_3H_6$) hajmiy ulushini (%) aniqlang.** | |
| **35. Aralashmani to‘liq yoqish uchun zarur havo hajmini (l, n.sh.; havoda 21% kislorod bor) aniqlang.** | |

**36.** $KMnO_4 + K_2SO_3 + H_2O \rightarrow MnO_2 + K_2SO_4 + KOH$ reaksiya tenglamasini tenglang va barcha koeffitsiyentlar **yig‘indisini aniqlang.**

**Javob:** _____

**Diqqat!** Javobingizni javoblar varaqasiga ko‘chirib yozing.

**37.** 50 g $CaCO_3$ reaksiyaga kirishdi ($CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$). Ajralgan $CO_2$ **hajmini (l, n.sh.) hisoblang.**

**Javob:** _____

**Diqqat!** Javobingizni javoblar varaqasiga ko‘chirib yozing.

**38.** 75,2 g $Cu(NO_3)_2$ qizdirilganda qisman parchalandi, qattiq qoldiq massasi 53,6 g bo‘ldi. Ajralgan gazlar (n.sh.) **hajmini (l) hisoblang.**

**Javob:** _____

**Diqqat!** Javobingizni javoblar varaqasiga ko‘chirib yozing.

**39.** Etilen va asetilendan iborat 5,6 l (n.sh.) aralashma bromli suvdan o‘tkazilganda 56 g brom yutildi. Aralashmadagi asetilenning **hajmini (l, n.sh.) toping.**

**Javob:** _____

**Diqqat!** Javobingizni javoblar varaqasiga ko‘chirib yozing.

**40.** 14,8 g $HCOOC_2H_5$ mo‘l NaOH bilan to‘liq gidrolizlandi. Hosil bo‘lgan spirt **massasini (g) aniqlang.**

**Javob:** _____

**Diqqat!** Javobingizni javoblar varaqasiga ko‘chirib yozing.

## YOZMA ISH

**41. Topshiriqni bajarish jarayonida barcha reaksiya tenglamalarini yozing. Miqdoriy va arifmetik hisoblashlarni (unga asos bo‘lgan usul yoki formulani ko‘rsatgan holda) uzviy ketma-ketlikda bajaring.**

Quyidagi eritmalar berilgan: HCl ning 0,02 mol/l li eritmasi va NaOH ning 0,01 mol/l li eritmasi. Ularni aralashtirib, umumiy hajmi 1 l bo‘lgan eritma tayyorlandi (0,3 l HCl eritmasi va 0,7 l NaOH eritmasi). Shuningdek, sirka kislotaning 0,1 mol/l li eritmasida dissotsilanish darajasi 1%.

1) Aralashtirilgandan keyingi eritmaning pH ini aniqlang.
2) Dastlabki HCl eritmasining pH ini hisoblang. 1 l shu eritmadagi HCl massasini (g) ham hisoblang.
3) Dastlabki NaOH eritmasining pH ini hisoblang.
4) Sirka kislota eritmasidagi $H^+$ ionlari konsentratsiyasini (mol/l) va pH ni aniqlang ($\lg 2 = 0{,}3$, $\lg 5 = 0{,}7$ ni ishlating). Shuningdek NaOH eritmasi 100 marta suyultirilsa pH qancha bo‘ladi?
5) Sirka kislotaning dissotsilanish konstantasini hisoblang ($K = C\alpha^2/(1-\alpha)$). Shuningdek HCl eritmasi 10 marta suyultirilsa pH necha bo‘ladi?

**Diqqat!** Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

**42.** Quyidagi o‘zgarishlar zanjirida $X$, $Y$, $Z$ noma’lum organik moddalar:

$$CH_3Cl \xrightarrow{Na} X \xrightarrow{Cl_2,\ \text{yorug‘lik}} Y \xrightarrow{KOH\ (\text{suvli})} Z \xrightarrow{CuO,\ t^\circ} CH_3CHO$$

1) $CH_3Cl \to X$ : reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.
2) $X \to Y$ : reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.
3) $Y \to Z$ : reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.
4) $Z \to CH_3CHO$ : reaksiya tenglamasini yozing; hosil bo‘lgan moddani nomlang.

**Diqqat!** Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

**43. Topshiriqni bajarish jarayonida barcha reaksiya tenglamalarini yozing. Miqdoriy va arifmetik hisoblashlarni (unga asos bo‘lgan usul yoki formulani ko‘rsatgan holda) uzviy ketma-ketlikda bajaring.**

Magniy (Mg) va rux (Zn) dan iborat 15,4 g qotishma mo‘l suyultirilgan sulfat kislotada eritildi, bunda 6,72 l (n.sh.) gaz ajraldi. Hosil bo‘lgan eritmaga mo‘l NaOH eritmasi qo‘shildi, hosil bo‘lgan cho‘kma ajratib olindi va havoda qizdirib quritildi.

1) Reaksiya tenglamalarini yozing va qotishmadagi har bir metallning massasini (g) aniqlang.
2) Qotishmadagi Mg ning massa ulushini (%) hisoblang.

3) Eritish uchun sarflangan 20% li $H_2SO_4$ eritmasining massasini (g) aniqlang.
4) Mo‘l NaOH qo‘shilganda hosil bo‘lgan cho‘kmaning formulasi va massasini (g) toping (reaksiya tenglamalarini yozing).
5) Cho‘kma havoda to‘liq qizdirilganda hosil bo‘lgan oksidning massasini (g) hisoblang.

**Diqqat!** Mazkur topshiriq kengaytirilgan javobni talab etib, uning yechilish usuli va shu usul asosidagi barcha hisoblash amallari javoblar varaqasida aks etishi lozim.

## 16-variant

**Javoblar kaliti**

*Faqat o‘qituvchi uchun.*

**1–32-savollar**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *B* | *C* | *D* | *D* | *D* | *C* | *A* | *C* | *D* | *B* | *A* | *C* | *D* | *A* | *C* | *C* |
| **17** | **18** | **19** | **20** | **21** | **22** | **23** | **24** | **25** | **26** | **27** | **28** | **29** | **30** | **31** | **32** |
| *A* | *C* | *B* | *C* | *A* | *A* | *D* | *D* | *D* | *B* | *B* | *B* | *A* | *B* | *A* | *B* |

**33–35-savollar**

| No | Javob | Qiymat |
|---|---|---|
| 33 | *E* | 1,12 |
| 34 | *D* | 14,29 |
| 35 | *A* | 88 |

**36–40-savollar**

| No | To‘g‘ri javob |
|---|---|
| 36 | 13 |
| 37 | 11,2 |
| 38 | 11,2 |
| 39 | 2,24 |
| 40 | 9,2 |

**41–43-topshiriqlar: javoblar**

**41-topshiriq.**

1) pH = 11
2) pH = 1,7; 0,73 g
3) pH = 12
4) pH = 3; 10
5) $1{,}01 \cdot 10^{-}5$; pH = 2,7

**42-topshiriq.**

1) $CH_3Cl + Na \rightarrow C_2H_6$ (etan)
2) $C_2H_6 + Cl_2$, yorug‘lik $\rightarrow C_2H_5Cl$ (xloretan)
3) $C_2H_5Cl + KOH$ (suvli) $\rightarrow C_2H_5OH$ (etanol)
4) $C_2H_5OH + CuO$, t° $\rightarrow CH_3CHO$ (sirka aldegid (etanal))

**43-topshiriq.**

1) 2,4 g; 13 g
2) 15,6%
3) 147 g
4) 5,8 g
5) 4 g

### Mavzular taqsimoti

| No | Mavzu | Tekshiriladigan ko‘nikma | Ball |
|---|---|---|---|
| 1 | Tirkashev VI bob (muvozanat, jadval) | Jadval tahlili | 1 |
| 2 | 11-sinf (Mendeleyev–Klapeyron) | Hisoblash va masala yechish | 1 |
| 3 | 8-sinf; Tirkashev II bob (kimyoviy bog‘lanish) | Bilim va tushunish | 1 |
| 4 | Tirkashev VI bob (tezlik, jadval) | Jadval tahlili | 1 |
| 5 | Tirkashev *V* bob; 11-sinf 7-bob | Bilim va tushunish | 1 |
| 6 | 7-sinf (Avogadro) | Hisoblash va masala yechish | 1 |
| 7 | 8-sinf; Tirkashev VI bob | Hisoblash va masala yechish | 1 |
| 8 | 11-sinf 4-bob (normal konsentratsiya) | Hisoblash va masala yechish | 1 |
| 9 | Tirkashev *V* bob; Tashev ($HNO_3$, $H_2SO_4$) | Hisoblash va masala yechish | 1 |

| | | | |
|---|---|---|---|
| 10 | Tirkashev VI bob (kimyoviy muvozanat) | Hisoblash va masala yechish | 1 |
| 11 | 9-sinf (gidroliz); Tirkashev IV bob | Bilim va tushunish | 1 |
| 12 | 7-sinf (kristallogidratlar) | Grafik tahlili | 1 |
| 13 | 11-sinf 8-bob; Tirkashev $V$ bob (elektroliz) | Bilim va tushunish | 1 |
| 14 | 9-sinf (genetik bog‘lanish) | Rasm bo‘yicha tahlil | 1 |
| 15 | 8-sinf; 9-sinf (genetik bog‘lanish) | Bilim va tushunish | 1 |
| 16 | 9-sinf; Tirkashev (mineral o‘g‘itlar/nodir gazlar, hisob) | Hisoblash va masala yechish | 1 |
| 17 | Tashev (nitratlar termik parchalanishi) | Hisoblash va masala yechish | 1 |
| 18 | 9-sinf; Tirkashev ($S/SO_2/H_2SO_4$, hisob) | Hisoblash va masala yechish | 1 |
| 19 | 9-sinf; Tirkashev ($N/NH_3/HNO_3$, hisob) | Hisoblash va masala yechish | 1 |
| 20 | 10-sinf (alkan, hisob) | Hisoblash va masala yechish | 1 |
| 21 | 10-sinf (ikki bosqichli hisob) | Hisoblash va masala yechish | 1 |
| 22 | 10-sinf (mavzuga oid paragraflar); Abdusamatov | Bilim va tushunish | 1 |
| 23 | 10-sinf (olinish usullari) | Bilim va tushunish | 1 |
| 24 | 10-sinf (mavzuga oid paragraflar); Abdusamatov | Bilim va tushunish | 1 |
| 25 | 10-sinf §6, §12, §20, §28 (izomeriya) | Bilim va tushunish | 1 |
| 26 | 10-sinf §28 (efirlanish) | Hisoblash va masala yechish | 1 |
| 27 | 10-sinf (ugl, hisob) | Hisoblash va masala yechish | 1 |
| 28 | 10-sinf (mavzuga oid paragraflar); Abdusamatov | Bilim va tushunish | 1 |
| 29 | 10-sinf (mavzuga oid paragraflar); Abdusamatov | Bilim va tushunish | 1 |
| 30 | Tirkashev III bob (eritmalar) | Hisoblash va masala yechish | 1 |
| 31 | 7-sinf (xavfsizlik qoidalari); 10-sinf lab. | Bilim va tushunish | 1 |
| 32 | 7-sinf (5-rasm) | Rasm bo‘yicha tahlil | 1 |
| 33–35 | Aralash masala (moslashtirish) | Ko‘p bosqichli hisob | 3 |
| 36–40 | Qisqa javobli masalalar | Hisoblash | 5 |
| 41–43 | Yozma ish | Usul va hisob (M, A) | 75 |